

Simulations

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Lasso-Cox

$$L(\boldsymbol{\beta}) = \prod_{r \in D} \frac{\exp(\boldsymbol{\beta}^T \mathbf{x}^{j_r})}{\{\sum_{j \in R_r} \exp(\boldsymbol{\beta}^T \mathbf{x}^j)\}} \quad : \text{Partial Likelihood}$$

$$\hat{\boldsymbol{\beta}} = \operatorname{argmin} \ell(\boldsymbol{\beta}), \quad \text{subject to } \sum |\beta_j| \leq s \quad (\text{Constraint form})$$

$$\hat{\boldsymbol{\beta}} = \operatorname{argmax}_{\boldsymbol{\beta}} \left\{ \ell(\boldsymbol{\beta}) - \lambda \sum_j |\beta_j| \right\} \quad (\text{Lagrangian form})$$

Reference: Tibshirani, R. (1997). The lasso method for variable selection in the Cox model. *Statistics in Medicine*, 16(4), 385–395.

Elastic net-Cox

We again consider the survival framework of Section 1. For the time being we assume no ties in failure/censoring time. We wish to find β which maximizes

$$L(\beta) = \prod_{i=1}^m \frac{e^{x_{j(i)}^\top \beta}}{\sum_{j \in R_i} e^{x_j^\top \beta}}$$

subject to our constraint: $\alpha \sum |\beta_i| + (1 - \alpha) \sum \beta_i^2 \leq c$. Maximizing the partial likelihood is equivalent to maximizing a scaled log partial likelihood,

$$\frac{2}{n} \ell(\beta) = \frac{2}{n} \left[\sum_{i=1}^m x_{j(i)}^\top \beta - \log \left(\sum_{j \in R_i} e^{x_j^\top \beta} \right) \right]$$

Reference: Simon, N., Friedman, J., Hastie, T., & Tibshirani, R. (2011). Regularization paths for Cox's proportional hazards model via coordinate descent. *Journal of Statistical Software*, 39(5), 1–13.

α : Hyperparameter

For fixed λ , as α changes from 0 to 1 our solutions move from more ridge-like to more lasso-like, increasing sparsity but also increasing the magnitude of all non-zero coefficients.

$$\hat{\beta} = \operatorname{argmax}_{\beta} \left[\frac{2}{n} \left(\sum_{i=1}^m x_{j(i)}^\top \beta - \log \left(\sum_{j \in R_i} e^{x_j^\top \beta} \right) \right) - \lambda P_{\alpha}(\beta) \right] \quad (1)$$

where,

$$\lambda P_{\alpha}(\beta) = \lambda \left(\alpha \sum_{i=1}^p |\beta_i| + \frac{1}{2} (1 - \alpha) \sum_{i=1}^p \beta_i^2 \right)$$

Grouped Lasso-Cox

Linear regression based ↗

$$\frac{1}{2} \left\| Y - \sum_{j=1}^J X_j \beta_j \right\|^2 + \lambda \sum_{j=1}^J \|\beta_j\|_{K_j},$$

$$K_j = p_j I_{p_j}$$

$$\hat{\beta} = \arg \max_{\beta} \left\{ \ell(\beta) - \lambda \sum_j \sqrt{p_j} \|\beta_j\|_2 \right\}$$

For a vector $\eta \in R^d$, $d \geq 1$, and a symmetric $d \times d$ positive definite matrix K , we denote

$$\|\eta\|_K = (\eta' K \eta)^{1/2}.$$

$$\|\beta_j\|_{K_j} = (\beta_j^T K_j \beta_j)^{1/2} = \sqrt{p_j} \|\beta_j\|_2$$

Reference:

- 1) Penalty: Yuan, M., & Lin, Y. (2006). Model selection and estimation in regression with grouped variables. *Journal of the Royal Statistical Society B*, 68(1), 49–67.
- 2) IRLS: Simon, N., Friedman, J., Hastie, T., & Tibshirani, R. (2011). Regularization paths for Cox's proportional hazards model via coordinate descent. *Journal of Statistical Software*, 39(5), 1–13.
- 3) Meier, L., van de Geer, S., & Bühlmann, P. (2008). The group lasso for logistic regression. *Journal of the Royal Statistical Society B*, 70(1), 53–71.

* IRLS Algorithm

2.2. Penalized least squares

We have reduced our problem to repeatedly solving the penalized, weighted least squares problem (2)

$$\hat{\beta} = \operatorname{argmin}_{\beta} \frac{1}{n} \sum_{i=1}^n w(\tilde{\eta})_i (z(\tilde{\eta})_i - x_i^T \beta)^2 + \lambda P_{\alpha}(\beta) \quad (3)$$

$$w(\tilde{\eta})_k = \ell''(\tilde{\eta})_{k,k} = \sum_{i \in C_k} \left[\frac{e^{\tilde{\eta}_k} \sum_{j \in R_i} e^{\tilde{\eta}_j} - (e^{\tilde{\eta}_k})^2}{\left(\sum_{j \in R_i} e^{\tilde{\eta}_j} \right)^2} \right]$$

$$z(\tilde{\eta})_k = \tilde{\eta}_k - \frac{\ell'(\tilde{\eta})_k}{\ell''(\tilde{\eta})_{k,k}} = \tilde{\eta}_k + \frac{1}{w(\tilde{\eta})_k} \left[\delta_k - \sum_{i \in C_k} \left(\frac{e^{\tilde{\eta}_k}}{\sum_{j \in R_i} e^{\tilde{\eta}_j}} \right) \right]$$

Sparse Grouped Lasso-Cox

$$\hat{\beta} = \operatorname{argmin}_{\beta} \frac{1}{n} \left[\log \left(\sum_{i \in D} \left(\sum_{j \in R_i} \exp(x_j^{\top} \beta) - x_i^{\top} \beta \right) \right) \right] \\ + (1 - \alpha) \lambda \sum_{l=1}^m \sqrt{p_l} \|\beta^{(l)}\|_2 + \alpha \lambda \|\beta\|_1,$$

where D is the set of failure indices, R_i is the set of indices, j , with $y_j \geq y_i$ (those still at risk at failure time i).

Reference: Simon, N., Friedman, J., Hastie, T., & Tibshirani, R. (2013). A sparse-group lasso. *Journal of Computational and Graphical Statistics*, 22(2), 231–245.

Metrics

1) C-index

$$H_c = \frac{CC + 0.5 \times TR}{CC + DC + TR}.$$

CC = concordant pairs, DC = discordant pairs, TR = tied-risk pairs
0.5 = random prediction, 1.0 = perfect prediction

Reference: Bertsimas, D., Dunn, J., Gibson, E., & Orfanoudaki, A. (2022). Optimal survival trees. *Machine Learning*, 111, 2951–3000.

Metrics

2) TPR

$$TPR = \frac{|\text{selected} \cap \text{true_active}|}{|\text{true_active}|}$$

Proportion of truly active features correctly recovered.

3) FPR

$$FDR = \frac{|\text{selected} - \text{true_active}|}{|\text{selected}|}$$

Proportion of selected features that are truly inactive.

Metrics

4) Network Coherence (Newman-Girvan Modularity)

So we further define the row (or column) sums $a_i = \sum_j e_{ij}$, which represent the fraction of edges that connect to vertices in community i . In a network in which edges fall between vertices without regard for the communities they belong to, we would have $e_{ij} = a_i a_j$. Thus we can define a modularity measure by

$$Q = \sum_i (e_{ii} - a_i^2) = \text{Tr } \mathbf{e} - \|\mathbf{e}^2\|, \quad (5)$$

where $\|\mathbf{x}\|$ indicates the sum of the elements of the matrix $\mathbf{x}$. This quantity measures the fraction of the edges

Measures how strongly selected features concentrate within the same graph module.

Higher value = more biologically coherent selection.

Reference: Newman, M.E.J., & Girvan, M. (2004). Finding and evaluating community structure in networks. *Physical Review E*, 69(2), 026113.

$e_{ii} \rightarrow e_m/c$:fraction of edges within community

$ma_i^2 \rightarrow (d_m/2c)^2$:squared fraction of edge endpoints in community m

$$Q = \sum_m \left(\frac{e_m}{c} - \left(\frac{d_m}{2c} \right)^2 \right)$$

where e_m = number of edges within community m ,

d_m = sum of degrees of nodes in community m ,

c = total number of edges in the graph

Metrics

5) Integrated Brier Score (IBS)

Brier Point Score:

$$BP_{\tau} = \frac{1}{|\mathcal{I}_{\tau}|} \sum_{i \in \mathcal{I}_{\tau}} (\hat{S}_i(\tau) - \mathbb{1}(t_i > \tau))^2,$$

where $\mathcal{I}_{\tau} = \{i \in \{1, \dots, n\}, |t_i \geq \tau \text{ or } \delta_i = 1\}$.

$$IB = \frac{1}{t_{max}} \frac{1}{n} \sum_{i=1}^n \int_0^{t_i} \frac{(1 - \hat{S}_i(t))^2}{\hat{G}(t)} dt + \delta_i \int_{t_i}^{t_{max}} \frac{(\hat{S}_i(t))^2}{\hat{G}(t)} dt.$$

Lower is better (0 = perfect, 0.25 = random).

Measures overall calibration of predicted survival probabilities.

Reference: Bertsimas, D., Dunn, J., Gibson, E., & Orfanoudaki, A. (2022). Optimal survival trees. *Machine Learning*, 111, 2951–3000.

- IPCW weight $\hat{G}(t)$: censoring distribution estimated via Kaplan-Meier
- Two interval integrals: $[0, t_i]$: pre-event interval $\rightarrow (1 - \hat{S}_i(t))^2$
- $[t_i, t_{max}]$: post-event interval $\rightarrow \hat{S}_i(t)^2$ only when event occurred, $\delta_i = 1$
- Normalized by $1/t_{max}$

Setting

$n=100$, $p=30$, train/test=70/30

fixed_lambda = 0.1 (same for all 4 methods)

[1] Lasso-Cox

- fixed_lambda = 0.1
- l1_ratio = 1.0 (pure ℓ_1)

[2] Elastic Net-Cox

- fixed_lambda = 0.1
- l1_ratio = 0.5 (equal ℓ_1/ℓ_2 mix)

[3] Group Lasso-Cox

- fixed_lambda = 0.1
- max_irls_iter = 20
- irls_tol = $1e-4$

[4] SGL-Cox

- fixed_lambda = 0.1
- alpha = 0.5 (equal ℓ_1 /group mix)
- max_irls_iter = 20
- irls_tol = $1e-4$

DGP

rho = 0.1 / 0.5

seed = 2026 ~ 2125 (100 replicates)

beta = (+0.5, +0.4, -0.3)

censoring = 30%

Result

No CV

$\rho=0.1$

```
=== Done! Total time: 47.1s ===  
Total rows: 400  
  
=== Summary ===  
      c_index      ibs      tpr      fdr      network_coherence  
      mean      std      mean      std      mean      std      mean      std      mean      std  
method  
ElasticNet_Cox  0.7242  0.0647  0.1796  0.0510  0.935  0.0757  0.5169  0.0591      0.4529  0.0797  
GroupLasso_Cox  0.5269  0.0723  0.1754  0.0656  0.086  0.2323  0.0880  0.2195      0.0322  0.1191  
Lasso_Cox       0.7111  0.0663  0.1806  0.0504  0.807  0.1208  0.3460  0.1150      0.3939  0.1158  
SGL_Cox         0.5732  0.0974  0.1845  0.0550  0.251  0.3157  0.2020  0.2424      0.0915  0.1739
```

$\rho=0.1$ (약한 상관): 변수 간 독립에 가까워 네트워크 구조가 무의미.
그룹 패널티가 방해로 작용 → GroupLasso TPR 0.086으로 붕괴.
Scenario 6(misspecified network)의 네거티브 컨트롤과 유사한 현상.

Result

No CV
rho=0.1

For MIQP

```
=== Done! Total time: 5493.4s ===  
Total rows: 100  
  
=== Summary ===  
      c_index    tpr    fdr  network_coherence  
mean    0.6857  0.6608  0.3392             0.3096  
std     0.0747  0.1133  0.1133             0.1351
```

lambda1=0.01

lambda2=0.01

gamma=0.1

Result

No CV

$\rho=0.5$

```
=== Done! Total time: 218.2s ===  
Total rows: 400
```

```
=== Summary ===
```

	c_index		ibs		tpr		fdr		network_coherence	
method	mean	std	mean	std	mean	std	mean	std	mean	std
ElasticNet_Cox	0.7903	0.0571	0.1378	0.0367	0.858	0.1093	0.4781	0.0749	0.4288	0.1028
GroupLasso_Cox	0.7249	0.0799	0.1620	0.0354	0.662	0.2228	0.6403	0.0712	0.3109	0.2009
Lasso_Cox	0.7816	0.0572	0.1398	0.0347	0.721	0.1351	0.3511	0.1298	0.3653	0.1436
SGL_Cox	0.7080	0.0838	0.1706	0.0393	0.605	0.1966	0.5633	0.0900	0.3069	0.1812

$\rho=0.5$:상관 강화 → 그룹 방법 TPR·coherence 회복, 단 FDR 통제 악화.

Lasso/ElasticNet은 상관 변수 중 대표 선택 전략으로 여전히 안정적. 구조 활용의 이득과 FDR 비용이 공존하는 상황.

Result

No CV
rho=0.5

For MIQP

```
=== Done! Total time: 4917.1s ===  
Total rows: 100
```

```
=== Summary ===
```

	c_index	tpr	fdr	network_coherence
mean	0.7315	0.5161	0.4839	0.3412
std	0.0719	0.1635	0.1635	0.1556

lambda1=0.01

lambda2=0.01

gamma=0.1

1. Strong joint, weak marginal effects.

- 개별 유전자 신호는 약하지만 (small Beta), pathway 전체가 함께 작용할 때 예후에 강한 영향
- 실제 암 생물학에서 흔한 상황 (예: PI3K-AKT pathway 내 여러 유전자의 누적 효과)
- Lasso는 개별 신호 강도에 의존 → 약한 신호 변수를 0으로 밀어버림
- MIQP의 graph reward는 연결된 변수를 함께 선택하려는 경향 → 약한 신호 변수가 이웃 변수의 선택에 끌려 같이 선택됨

2.5 Discrete Optimization Formulation

The proposed optimization problem is

$$\min_{\beta, s, \xi} \sum_{(i,j) \in \mathcal{P}} \xi_{ij}^2 + \lambda_2 \sum_{j=1}^p \beta_j^2 + \lambda_1 \sum_{j=1}^p s_j - \gamma \sum_{(j,k) \in E} w_{jk} s_j s_k \quad (2.5)$$

$$\text{subject to } \xi_{ij} \geq 1 - \sum_{j=1}^p \beta_j (X_{ij}^\Delta)_j, \quad (i,j) \in \mathcal{P}, \quad (2.6)$$

$$\xi_{ij} \geq 0, \quad (i,j) \in \mathcal{P}, \quad (2.7)$$

$$-Ms_j \leq \beta_j \leq Ms_j, \quad j = 1, \dots, p, \quad (2.8)$$

$$s_j \in \{0, 1\}, \quad j = 1, \dots, p, \quad (2.9)$$

$$\sum_{j=1}^p s_j \leq K. \quad (2.10)$$

$$-\gamma \sum w_{jk} s_j s_k \text{ (Reward)}$$

그래프에서 연결된 두 변수 j와 k가 둘 다 선택($s_j=1, s_k=1$) 되면

$-\gamma w_{jk}$ 만큼 목적함수 값이 낮아짐

→ 낮아질수록 좋은 해

→ 연결된 변수들을 함께 선택하는 방향으로 유도

Suggestion

module 1 $\beta_1 = +0.5$

module 2 $\beta_2 = +0.4$

module 3 $\beta_3 = -0.3$



Small Beta?
Another lambda?

Why Lasso fails in weak marginal signal

① KKT Conditions

$$|X_j^T(Y - X\hat{\beta})|/n \leq \lambda \Rightarrow \hat{\beta}_j = 0 \quad \longleftarrow \quad \text{변수 } j \text{의 설명력을 확인하는 조건}$$

변수 j 가 선택되려면 잔차와의 내적, 즉 marginal correlation이 λ 보다 커야 함, 신호가 약하면 조건 만족 x.

"Lasso는 marginal 신호 강도에 의존한다"

Lemma 2.1. Denote the gradient of $n^{-1}\|\mathbf{Y} - \mathbf{X}\beta\|_2^2$ by $G(\beta) = -2\mathbf{X}^T(\mathbf{Y} - \mathbf{X}\beta)/n$.
Then a necessary and sufficient condition for $\hat{\beta}$ to be a solution of (2.2) is:

$$\begin{aligned} G_j(\hat{\beta}) &= -\text{sign}(\hat{\beta}_j)\lambda \text{ if } \hat{\beta}_j \neq 0, \\ |G_j(\hat{\beta})| &\leq \lambda \text{ if } \hat{\beta}_j = 0. \end{aligned}$$

Moreover, if the solution of (2.2) is not unique (e.g. if $p > n$) and $G_j(\hat{\beta}) < \lambda$ for some solution $\hat{\beta}$, then $\hat{\beta}_j = 0$ for all solutions of (2.2).

② Beta min Condition

$$|\beta^0|_{\min} := \min_{j \in S_0} |\beta_j^0|,$$

condition requiring some lower non-zero bound on $|\beta^0|_{\min}$ a “beta-min” condition

$\rho_n := \min_{i \in S} |\beta_i^*|$, and the sparsity index $s = |S|$.

$$\frac{1}{\rho_n} \left[\lambda_n + \sqrt{\frac{\log s}{n}} \right] \rightarrow 0,$$

active set 내 가장 작은 계수의 절댓값

$n_{\text{train}} = 70, s = 10, \lambda = 0.1$ in our simulation

0.937 $\rightarrow$ closer to 1

Beta를 더 작게 해서 비율이 1을 넘으면 Lasso가 실패할 가능성이 높아지는 것

Bühlmann, P., & van de Geer, S. (2011). Statistics for High-Dimensional Data. Springer.

Wainwright, M. J. (2009). Sharp thresholds for high-dimensional and noisy sparsity recovery using ℓ_1 -constrained quadratic programming (Lasso). IEEE Transactions on Information Theory, 55(5), 2183–2202.

2. Correlated pathway genes.

For $\rho=0,1$

TPR

방법	$\rho=0.1$	$\rho=0.5$
Lasso_Cox	0.807	0.721
ElasticNet_Cox	0.935	0.858
GroupLasso_Cox	0.086	0.662
SGL_Cox	0.251	0.605

그룹 구조 신호가 약함, GroupLasso/SGL



$\rho=0.1$ (약한 상관) 환경에서 MIQP는 TPR 0.661로, GroupLasso(0.086)보다 우수, ElasticNet(0.935)보다는 낮음.

MIQP의 graph co-selection reward는 모듈 내 변수 간 상관이 강할 때 ($\rho=0.5$) 효과적

($\rho=0.1$ 처럼 상관이 약한 상황에서는 개별 변수의 marginal 신호에 의존하는 ElasticNet도 성능 좋음)

```
=== Done! Total time: 5493.4s ===
Total rows: 100

=== Summary ===
      c_index    tpr    fdr  network_coherence
mean    0.6857  0.6608  0.3392             0.3096
std     0.0747  0.1133  0.1133             0.1351
```

$\rho=0.1 \rightarrow$ 모듈 내 변수들이 서로 거의 독립 $\rightarrow$ 각 변수가 독립적으로 신호를 y 에 전달.
marginal 신호만 봐도 active 변수 포착 가능(Group structure 필요성 감소)

For $\rho=0,5$

TPR

방법	$\rho=0.1$	$\rho=0.5$
Lasso_Cox	0.807	0.721
ElasticNet_Cox	0.935	0.858
GroupLasso_Cox	0.086	0.662
SGL_Cox	0.251	0.605

상관이 강할수록 구조 기반 방법이 유리해짐

```
=== Done! Total time: 4917.1s ===  
Total rows: 100  
  
=== Summary ===  
      c_index   tpr   fdr  network_coherence  
mean   0.7315 0.5161 0.4839             0.3412  
std    0.0719 0.1635 0.1635             0.1556
```

Result

1> C-index (higher is better, 0.5 = random prediction)

방법	rho=0.1	rho=0.5
Lasso_Cox	0.711	0.782
ElasticNet_Cox	0.724	0.790
GroupLasso_Cox	0.527	0.725
SGL_Cox	0.573	0.708

C-index improves overall under rho=0.5 (higher correlation). ElasticNet and Lasso rank highest across both scenarios.

2> TPR (higher is better, proportion of true active features correctly recovered)

방법	rho=0.1	rho=0.5
Lasso_Cox	0.807	0.721
ElasticNet_Cox	0.935	0.858
GroupLasso_Cox	0.086	0.662
SGL_Cox	0.251	0.605

ElasticNet dominates at rho=0.1.
GroupLasso and SGL improve substantially at rho=0.5 — suggesting group structure becomes more exploitable under higher feature correlation.

Result

3> FDR (lower is better, proportion of falsely selected features)

방법	rho=0.1	rho=0.5
Lasso_Cox	0.346	0.351
ElasticNet_Cox	0.517	0.478
GroupLasso_Cox	0.088	0.640
SGL_Cox	0.202	0.563

GroupLasso shows lowest FDR at $\rho=0.1$, but TPR is also very low (0.086) — indicating it selects almost nothing.
GroupLasso FDR spikes at $\rho=0.5$.

4> Network Coherence (Higher is better)

방법	rho=0.1	rho=0.5
Lasso_Cox	0.394	0.365
ElasticNet_Cox	0.453	0.429
GroupLasso_Cox	0.093	0.311
SGL_Cox	0.092	0.307

ElasticNet ranks highest in both scenarios.
GroupLasso and SGL improve at $\rho=0.5$.

Result

5> IBS(lower is better)

방법	$\rho=0.1$	$\rho=0.5$
Lasso_Cox	0.1806	0.1398
ElasticNet_Cox	0.1796	0.1378
GroupLasso_Cox	0.1754	0.1620
SGL_Cox	0.1845	0.1706

All four methods show similar performance levels.

lambda1=0.005, lambda2=0.005, gamma=0.1 for MIQP

$\lambda=0.05$ 가 TPR은 높이지만 FDR도 같이 올라가는 trade-off

No CV
rho=0.1

```
=== Done! Total time: 605.8s ===
```

```
Total rows: 400
```

```
=== Summary ===
```

method	c_index mean	std	ibs mean	std	tpr mean	std	fdr mean	std	network_coherence mean	std
ElasticNet_Cox	0.7217	0.0642	0.1877	0.0525	0.967	0.0570	0.6045	0.0357	0.4713	0.0520
GroupLasso_Cox	0.6399	0.0798	0.1949	0.0424	0.856	0.1766	0.6615	0.0151	0.4355	0.0947
Lasso_Cox	0.7240	0.0643	0.1822	0.0516	0.931	0.0761	0.5197	0.0578	0.4534	0.0784
SGL_Cox	0.6258	0.0779	0.1966	0.0423	0.707	0.1719	0.6153	0.0601	0.3921	0.1334

TPR이 전반적으로 크게 증가 — λ 줄이니까 더 많은 변수 선택
GroupLasso, SGL 개선: TPR 0.086→0.856, 0.251→0.707
(FDR도 함께 증가 — 변수를 더 많이 선택하다 보니 오선택 증가)
c_index는 소폭 변화 — 예측 성능 자체는 크게 안 변함

rho=0.5

```
=== Done! Total time: 1096.7s ===
```

```
Total rows: 400
```

```
=== Summary ===
```

method	c_index mean	std	ibs mean	std	tpr mean	std	fdr mean	std	network_coherence mean	std
ElasticNet_Cox	0.7854	0.0569	0.1428	0.0401	0.890	0.1000	0.5681	0.0523	0.4567	0.0695
GroupLasso_Cox	0.6636	0.0920	0.1835	0.0380	0.846	0.2062	0.6547	0.0681	0.4254	0.1197
Lasso_Cox	0.7876	0.0566	0.1402	0.0384	0.826	0.1284	0.4691	0.0860	0.4190	0.1057
SGL_Cox	0.6578	0.0864	0.1870	0.0383	0.654	0.1817	0.6353	0.0800	0.4052	0.1215

TPR 소폭 증가

GroupLasso c_index 하락: 0.7249 → 0.6636 — λ 너무
줄이니 오히려 과선택으로 예측력 저하
FDR 증가 — rho=0.5에서도 동일 패턴
ibs 소폭 증가 — λ 줄이면서 모델 복잡도 올라간 영향

rho=0.5

For MIQP

```
=== Done! Total time: 5102.2s ===  
Total rows: 100  
  
=== Summary ===  
      c_index    tpr    fdr  network_coherence  
mean    0.7291  0.5290  0.4710             0.3413  
std     0.0699  0.1532  0.1532             0.1379
```

→ MIQP는 λ 변화에 둔감 — 변수 선택이 λ 보다 graph reward γ 에 의존하기 때문

→ Baseline(GroupLasso TPR 0.086→0.856)의 극적 변화와 대조적

Appendix

KKT condition for Elastic net

Zou, H., & Hastie, T. (2005) "Regularization and variable selection via the elastic net" *Journal of the Royal Statistical Society Series B*, 67(2), 301–320

Theorem 1. Given data $(\mathbf{y}, \mathbf{X})$ and parameters (λ_1, λ_2) , the response $\mathbf{y}$ is centred and the predictors $\mathbf{X}$ are standardized. Let $\hat{\beta}(\lambda_1, \lambda_2)$ be the naïve elastic net estimate. Suppose that $\hat{\beta}_i(\lambda_1, \lambda_2) \hat{\beta}_j(\lambda_1, \lambda_2) > 0$. Define

$$D_{\lambda_1, \lambda_2}(i, j) = \frac{1}{|\mathbf{y}|_1} |\hat{\beta}_i(\lambda_1, \lambda_2) - \hat{\beta}_j(\lambda_1, \lambda_2)|;$$

then

$$D_{\lambda_1, \lambda_2}(i, j) \leq \frac{1}{\lambda_2} \sqrt{2(1 - \rho)},$$

where $\rho = \mathbf{x}_i^T \mathbf{x}_j$, the sample correlation.

A.2. Proof of theorem 1

If $\hat{\beta}_i(\lambda_1, \lambda_2) \hat{\beta}_j(\lambda_1, \lambda_2) > 0$, then both $\hat{\beta}_i(\lambda_1, \lambda_2)$ and $\hat{\beta}_j(\lambda_1, \lambda_2)$ are non-zero, and we have $\text{sgn}\{\hat{\beta}_i(\lambda_1, \lambda_2)\} = \text{sgn}\{\hat{\beta}_j(\lambda_1, \lambda_2)\}$. Because of equation (4), $\hat{\beta}(\lambda_1, \lambda_2)$ satisfies

$$\left. \frac{\partial L(\lambda_1, \lambda_2, \beta)}{\partial \beta_k} \right|_{\beta = \hat{\beta}(\lambda_1, \lambda_2)} = 0 \quad \text{if } \hat{\beta}_k(\lambda_1, \lambda_2) \neq 0. \quad (17)$$

Hence we have

$$-2\mathbf{x}_i^T \{\mathbf{y} - \mathbf{X}\hat{\beta}(\lambda_1, \lambda_2)\} + \lambda_1 \text{sgn}\{\hat{\beta}_i(\lambda_1, \lambda_2)\} + 2\lambda_2 \hat{\beta}_i(\lambda_1, \lambda_2) = 0, \quad (18)$$